

# It's about time: moving away from statistical analysis of ecotoxicity data

Tjalling Jager, DEBtox Research, Stevensweert, The Netherlands

Email: [tjalling@debtox.nl](mailto:tjalling@debtox.nl)

Version: 7 November 2024

This preprint is not peer reviewed and made available at

[www.debtox.info/downloads/my\\_papers/Jager\\_about\\_time.pdf](http://www.debtox.info/downloads/my_papers/Jager_about_time.pdf)

This work is licensed under CC BY-NC-ND 4.0. To view a copy of this license, visit

<https://creativecommons.org/licenses/by-nc-nd/4.0/>

## Abstract

Environmental risk assessment (ERA) relies on laboratory testing to establish the toxic properties of a compound. However, ERA is not concerned with toxicity under laboratory conditions: it needs to assess the impacts of the compound in the real world. Data-driven statistical analyses (e.g., hypothesis testing and interpolation) are the common approaches for analysing toxicity data, but statistics is the wrong tool for the job at hand. ERA does not need a statistical description of the effects in the toxicity test (at the end of the standardised test duration), it needs to extrapolate from the lab test to longer and time-varying exposure. Extrapolation requires mechanistic process models, providing a simplified representation of the mechanisms underlying toxicity. Any useful model for the toxicity process should explicitly consider both dose (e.g., exposure concentration) and time. In the history of effects analysis for ERA, the factor of time does not get as much attention as the dose, hence common use of the term ‘dose-response analysis’. However, this is a historical oversight: time is a crucial factor for understanding toxicity and thereby essential for meaningful extrapolation from lab to field. Mechanistic models for ecotoxicity, considering both dose and time, have been around for quite some time and are classified as toxicokinetic-toxicodynamic (TKTD) models. TKTD models are starting to find their way into the pesticide ERA in Europe, next to the classical statistical approaches. In this opinion paper, I argue that it is about time to leave statistical analysis of toxicity data behind us. Statistics remains important for ERA’s effects assessment, but its role lies in the definition of appropriate ‘error models’, explaining the deviations between model output and observations, which is essential for parameter estimation, uncertainty quantification, and error propagation. The ‘process model’, essential for extrapolation, firmly belongs to TKTD modelling.

## INTRODUCTION

If the only tool you have is a hammer, you tend to see every problem as a nail. A hammer is a very useful tool, but clearly not the best choice for every problem. In this opinion paper, I will argue that statistical analysis is the wrong tool for the analysis of ecotoxicity data, particularly within the context of environmental risk assessment (ERA).

Laboratory toxicity testing is a mainstay of ERA, but what question are we addressing with these tests? Clearly, these toxicity tests are not meant to assess the chemical’s risk in the laboratory setting; they are meant to say something about the expected impacts of a chemical’s use to (populations of) organisms in the outside world. Therefore, this task requires methods to extrapolate from the effects observed in the laboratory test to potential effects under field-relevant conditions. This is not a trivial task: exposure in the field is always of a different duration than in the laboratory, and the exposure pattern (i.e., exposure concentration as a function of time) may be wildly different from the (constant) exposure pattern in the lab. Furthermore, environmental conditions (e.g., temperature) will be different, and even time-varying, in the field, and we may want to protect different species, or different life stages, than those tested in the lab.

The calculation methods for supporting ERA’s effect assessment are commonly referred to as ‘statistical analysis of ecotoxicity data’ (OECD 2006, and the title of this special series). At this point, it is important to clarify terminology. I view ‘statistical analysis’ as a calculation method, simple or complex, that is principally data-driven. No assumptions are made about the mechanisms that have led to the particular set of observations. The data points are viewed as random samples from a distribution, whose mean follows from the observations themselves or from a quite meaningless descriptive model (e.g., linear or log-logistic). In contrast, I will use the term ‘mechanistic models’ for

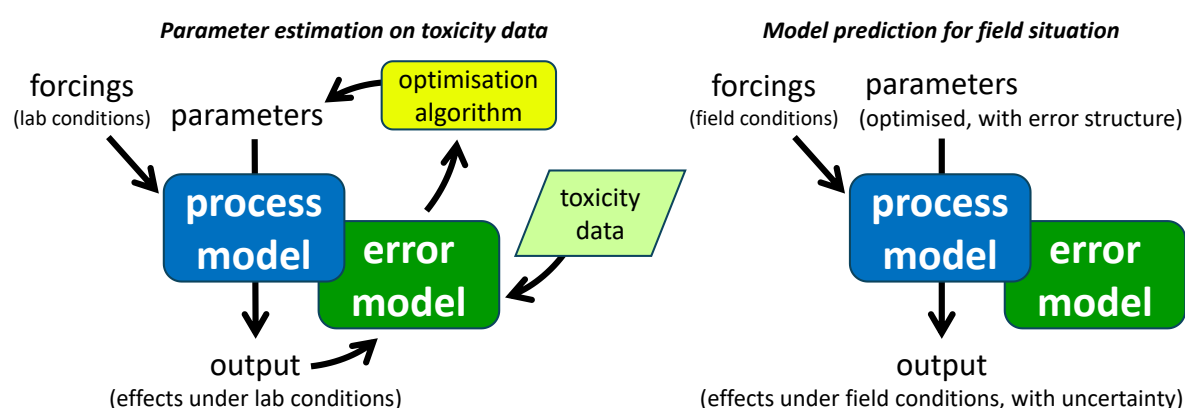
models that follow a theory-driven, or first-principles, approach, applying a simplified representation of reality that is based on biology, chemistry and physics, rather than on data. Mechanistic models should be formulated as a set of simplifying assumptions translated into (systems of) differential equations. The output of such a model will generally be deterministic, so combination of statistical methods and mechanistic models is common in practical applications. However, I do not classify such combined models under the term ‘statistical analysis’.

The classical methods for the analysis of ecotoxicity data rely heavily on the statistical toolboxes for hypothesis testing and interpolation. The goal of these tools is to arrive at a single number, specifying the toxicity of a chemical to a species. Hypothesis testing is used to generate metrics such as the NOEC (no-observed effect concentration; the highest exposure treatment without significant effects compared to the control, at the end of the test), and interpolation using a descriptive curve is used to yield metrics such as the EC<sub>x</sub> (the estimated concentration to yield *x*% effect, relative to the control, at the end of the test). However, what question have we answered with such a simple summary statistic? We now have a description of the effects at one time point (generally: the end of the toxicity test), for one specific exposure pattern (generally: constant), under the (constant) conditions of the lab test. There is no scientifically meaningful way to extrapolate such summary statistics to a field situation.

Clearly, statistics, used in isolation, is the wrong tool for ERA’s extrapolation problem; it can only make statements about the exact same situation as that in which the data set was generated (the end of a specific toxicity test in the laboratory). In the early days of ERA, using the proverbial hammer was preferable over not doing ERA at all, but it is about time to move forward and remedy this unfortunate situation. It should be obvious that the ERA question at hand is not of a statistical nature at all. Extrapolation and prediction are first and foremost modelling questions; more specifically, they require a mechanistic process model. As soon as we confront such a model with observations, we will need statistics as well, but the mechanistic model for the toxicity process should be our primary concern. In the remainder, I will take a closer look at the extrapolation problem, and the methods to deal with it properly.

## THE EXTRAPOLATION PROBLEM

Let’s take a step back and take an engineer’s perspective of the extrapolation problem in ERA. The challenge can be divided into two steps: the extraction of relevant (mechanistic) information from the laboratory toxicity data, and subsequently the use of that information to predict the toxic effects under the field conditions that we are interested in. This logical breakdown of the problem is illustrated in Figure 1. Note that I explicitly delineate two models: a model for the toxicity process and a model for the ‘error’ (i.e., the inevitable deviations between model output and the observations).



**Figure 1.** An engineer’s view of the regulatory challenge of extrapolating effects from laboratory toxicity data to the field. Note that the error model may also have parameters that need to be estimated from the data set (not shown in the schemes).

The process model has parameters, which together with the process model and the forcings (the exposure pattern and the environmental conditions) determines the model output. The output needs to

be of the same type as the observations (the toxicity data) so they can be compared. In parameter estimation, an optimisation algorithm tries a set of parameter values, the error model evaluates quantitatively to what extent the model output is a good representation of the observations, and informs the optimisation algorithm, which selects new parameter values to try to improve the correspondence between model and data. This cycle continues until the fit is judged to be ‘as good as it gets’. The error model should now enable us to present optimal parameter values with their associated uncertainty structure, given the data.

In the prediction step, the optimal parameter values with their uncertainty structure can subsequently be used to make predictions for the effects that can be expected under field conditions, again with an uncertainty estimate. In principle, the process model will stay the same, but the forcings will be different (in particular the exposure pattern). However, we may modify or extend the process model to allow prediction of different endpoints than those observed in the laboratory test (e.g., extrapolating to a different species or scale up to population dynamics).

What can we learn from this breakdown? Firstly, we should start by defining a suitable process model for the task at hand. Only after that should we consider an appropriate error model, based on the nature of the deviations between model prediction and the observations. It is important to stress that a lousy process model with a fantastically advanced error model will still produce nonsense. The reverse, a suitable process model with a poor error model, may still be quite useful for ERA, even though we should treat the resulting confidence intervals with care. Another thing we can learn from this breakdown is that statistical approaches are myopic: they are only concerned with the analysis of the toxicity data (analogous to the left panel of Fig. 1) and do not address the predictions that are needed in ERA. They can therefore only be considered as partial methods for effects assessment.

## DEMANDS ON THE PROCESS MODEL

What demands should we place on a process model? Its output should obviously match the observations from the laboratory toxicity test. In ERA, the most common tests focus on life-history traits such as survival, growth and reproduction, since the aim is to protect populations rather than the health of individual organisms. The toxic effects on life-history traits clearly depend on the dose (or exposure concentration), hence the commonly-used term ‘dose-response analysis’. However, dose is not the only important factor determining the toxicity: the other major factor is time. It is well known that toxicity depends on time (e.g., Sprague 1969), but it is important to realise that this dependence varies with the toxicant, the species, and the trait that is followed (e.g., Alda Álvarez et al. 2006). In some cases, effects appear rapidly after the onset of exposure, sometimes effects develop very slowly and may even continue long after the standard test duration, even when exposure has long ceased (see Jager 2024, for an example). Therefore, focusing on the effects after a standardised exposure time makes no sense.

Any process model for toxic effects that only considers dose and ignores time (or vice versa) is a lousy model. It is not going to be of any help in predicting effects under different (i.e., field-relevant) conditions. Therefore, we must move away from dose-response (and time-response) analysis and embrace dose-time-response analysis. In many regulatory frameworks, a further demand on the process model is that it should be able to deal with time-varying forcings, in particular: time-varying exposure patterns. In pesticide risk assessments, for example, fate models predict field-relevant exposure for different application scenarios that generally comprises complex patterns for the environmental concentrations over time. Experimentally testing the toxicity for all these patterns in the laboratory is highly impractical (and unethical), so we need a model framework that extrapolates across exposure patterns, and can thereby act as a virtual toxicity test (Sherborne et al. 2020).

Classical dose-response analysis applies tools from the statistical toolbox to analyse the data at the end of the laboratory toxicity test. Since these methods are descriptive, they can only make statements about the laboratory test (i.e., interpolation). ERA requires extrapolation, but this is outside of the scope of the statistical methods. Therefore, the results of the dose-response analysis have to be used in cruder ways. In pesticide risk assessment, for example, the detailed time-varying output of an advanced fate model is unceremoniously reduced to a single number (e.g., a peak or time-weighted average concentration) so that it can be compared to the NOEC or EC<sub>x</sub> (with an assessment factor). In essence, this train of calculations should be viewed as a model (analogous to the right panel of Fig. 1), although it is not treated as such. Every model follows from a set of simplifying assumptions about

the real world. If we would make the assumptions underlying these methods for effects assessment explicit, I doubt that anyone would be willing to defend them as suitable for ERA. To judge the usefulness of any approach for analysis of toxicity data, we should not forget the ‘model’ needed to produce the risk estimates. From the above, it should be clear that statistical methods cannot operate as process models for effects assessment, as outlined in Figure 1.

The answer to ERA’s extrapolation question is to use mechanistic models as the process model in Figure 1, and restrict statistics to the error model. Mechanistic models follow from explicit and meaningful assumptions about the underlying processes that lead to the toxicity, and are thereby able to support extrapolation. Please note that mechanistic modelling is already very common in ERA for other parts of the risk assessment: since its early days, ERA has been using fate models and toxicokinetic models. Nobody in their right mind would propose to use statistical methods instead of these mechanistic models. Therefore, there is plenty of experience with modelling and the use of model predictions in ERA, just not for toxic effects.

Table 1. Criteria for methods to be used in the effects assessment of environmental risk assessment. These criteria are relevant when quantitative statements about effects in the environment need to be derived from the results of laboratory toxicity tests.

| No.                                     | Topic                                       | Evaluation criterium                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Analysis of toxicity data</b>        |                                             |                                                                                                                                                                                                                                                                                                                                           |
| 1                                       | Dose and time                               | Does the method consider both exposure and time in an integrated analysis of the data from the toxicity test? This implies dose-time-response analysis of the toxicity data.                                                                                                                                                              |
| 2                                       | Time-varying exposure                       | Is the method able to use data from toxicity tests where exposure was not constant? The results should have the same interpretation as an analysis on data from a test with constant exposure.                                                                                                                                            |
| 3                                       | Multiple endpoints                          | If there are multiple endpoints in a toxicity test (e.g., growth and reproduction), does the method use all data in an integrated analysis?                                                                                                                                                                                               |
| 4                                       | Other environmental conditions              | Does the method offer handles to consider other factors in the data analysis? E.g., dealing with different or time-varying temperatures and food levels.                                                                                                                                                                                  |
| <b>Predictions for field conditions</b> |                                             |                                                                                                                                                                                                                                                                                                                                           |
| 5                                       | Extrapolate across time                     | Is the method able to produce meaningful extrapolations to longer time scales than those of the laboratory test? I.e., from short-term to long-term exposure.                                                                                                                                                                             |
| 6                                       | Extrapolate across exposure scenarios       | Is the method able to produce meaningful extrapolations to other exposure scenarios? E.g., from constant to time-varying exposure.                                                                                                                                                                                                        |
| 7                                       | Extrapolate across environmental conditions | Does the method offer handles to consider other factors in extrapolation? E.g., dealing with different or time-varying temperatures (or from individuals to populations).                                                                                                                                                                 |
| <b>Model evaluation</b>                 |                                             |                                                                                                                                                                                                                                                                                                                                           |
| 8                                       | Uncertainty analysis                        | Is the method capable of making predictions with an associated measure of uncertainty? This involves propagating the uncertainty from the analysis of the toxicity test data to predictions.                                                                                                                                              |
| 9                                       | Reliability                                 | Is the method corroborated for the intended purpose? In most ERA frameworks, the purpose implies extrapolation of the toxicity test results across time and across exposure scenarios. Corroboration involves comparison of model predictions with independent data and patterns that were not used in model development and calibration. |

In Table 1, I summarise the demands that we can place on the methods for effects assessment in ERA. Some answers may not have an absolute yes or no for a given approach, but it should be possible to attach a graded score. These points may be weighed differently for different ERA frameworks. For example, extrapolation from constant to time-varying exposure will be more important for pesticides than for a component of a detergent used by consumers. I would consider

criterion 1 and 5 to be crucial for any method for effects assessment in ERA. Criterion 2 and 6 become essential as soon as exposure in the lab and in the field can be time varying. Methods that fail these criteria cannot be repaired: they simply lack fundamental properties. Point 9 should also be very important for application in ERA, but if the corroboration effort for a method is deemed insufficient, it can be addressed by further research. A more detailed and technical series of tests for distinguishing between useful and useless models, underlying several of the criteria in the table, is provided by Kooijman (2018).

## **TKTD MODELS AS PROCESS MODELS**

The process models that are up to the task at hand should (probably by definition) be classified as toxicokinetic-toxicodynamic (TKTD) models. These models use a simplified representation of the uptake and elimination of the toxicant by the organism (TK), linked to a simplified representation of the organism, as far as relevant for the endpoint of interest (TD). Practical TKTD models for analysing ecotoxicity data have been around at least since the 1990's (Bedaux and Kooijman 1994; Kooijman and Bedaux 1996), and have since developed into more comprehensive frameworks for survival (Jager et al. 2011; Jager and Ashauer 2023) and sub-lethal effects (Jager 2020; Sherborne et al. 2020). The early TKTD models were already included in the 2006 guidance of the Organisation for Economic Co-operation and Development (OECD) on analysis of toxicity data under the heading 'biology-based methods' (OECD 2006). More recently, the European Food Safety Authority (EFSA) released a scientific opinion on the use of TKTD models for aquatic risk assessment of pesticides in Europe (EFSA 2018), and included TKTD models for survival into the revised guidance for risk assessment of bees (EFSA 2023).

Even though I would consider the current state-of-the-art TKTD models to be 'fit for purpose', and vastly superior to the classical dose-response methods for the questions of ERA, they are not beyond criticism and much work remains to be done. Obviously, the real world is always going to be more complex than the model (otherwise it would not be a model), but there is a limit to the amount of complexity that we can include into the model, owing to the type and extent of information that can be obtained from commonly-available data sets. The current TKTD models were designed to operate with the results from rather standard laboratory toxicity tests (following the effects on one or more traits over time), and nothing more. This amount of information will not be sufficient in all cases, for example when a parent compound and its metabolites all contribute (non-linearly) to the overall toxicity. It is good to stress that the current standard test protocols have been optimised for statistical dose-response analysis, and not for TKTD modelling. Reconsideration of the test protocols is therefore needed as well (Barsi et al. 2014; Jager 2014). More work is also needed to develop and test TKTD models for endpoints where no models are currently established (e.g., sex ratio and behavioural endpoints).

For ERA purposes, it makes sense to provide guidelines and standards for the use of models. In 2014, EFSA released a scientific opinion on good modelling practice for mechanistic effects models (EFSA 2014), which applies to models at the individual level (TKTD) but seems particularly concerned with population models. In general, this document makes reasonable recommendations. However, its conclusions should not only be applied to mechanistic effects models. I would make a plea to apply the same evaluation to non-mechanistic effects models (i.e., classical dose-response analysis, and how the results are subsequently compared to fate model output). These are models as well, albeit rather implicit ones. Even though TKTD models may not tick all the boxes of 'good modelling practice' yet (and some criteria may need to be applied differently, Jager and Ashauer 2018), I am confident that they will score much better on such an evaluation than the classical non-mechanistic methods.

## **DEMANDS ON THE ERROR MODEL**

For the error model, we do need to draw heavily from the statistical toolbox. A debate that is drawing quite a bit of attention in applied sciences is that between frequentist and Bayesian statistical frameworks. However, this debate is, in my opinion, of minor concern for our topic. Both frameworks have practical advantages and disadvantages but, when similar assumptions go into the analysis, both approaches will generally yield very similar results. Both frameworks usually rely on construction of a likelihood function for the parameters, given the data set. For survival data (as function of exposure

pattern and time), there is a rather straightforward application of the multinomial distribution (Bedaux and Kooijman 1994). However, for sub-lethal effects, such as those on growth and reproduction, defining an appropriate likelihood function turns out to be more problematic. In most cases, the deviations between model and data are pragmatically treated as random and independent measurement errors. However, observations are often not independent (e.g., when following the same individuals over time), we may have observations on multiple traits from the same individuals (e.g., growth and reproduction), and measurement error is not the most important cause of the deviations. More important causes of deviation are the fact that the model is wrong and that the individuals in the test are not identical. Some of these issues, and possible way forward, are discussed elsewhere (e.g., Jager 2013; Marques et al. 2019), but much more work is needed on this topic.

As I already stated, a good process model with a poor error model may still be quite useful in ERA, even though the confidence intervals need to be treated with care. In this respect, it is interesting to consider the role of uncertainty quantification in modern-day ERA. Most ERA frameworks are deterministic, so uncertainties are not quantified nor propagated to risk estimates. EFSA places a lot of emphasis on uncertainty analysis in their recent publications on mechanistic modelling for effects assessment (EFSA 2014; EFSA 2018). This is laudable; after all, the definition of risk embodies the probability of negative impacts. However, it is remarkable that the request for uncertainty quantification and propagation seems restricted to mechanistic effect models. No such requirements are formulated for the classical non-mechanistic effects analysis, nor for fate models. This imbalance needs to be addressed as well.

## CONCLUDING REMARKS

Statistical analysis is the wrong tool for effect assessment in ERA, like the proverbial hammer for driving in screws. ERA is not about describing and summarising the effects in the standardised laboratory test; the key question is the prediction of effects under field-relevant exposure conditions. Data-driven methods, however advanced, are not going to help us extrapolate from the laboratory test to the field; we need theory-driven process models. Therefore, it is time to move forward and replace the descriptive dose-response analysis by mechanistic dose-time-response modelling. In ERA of pesticides in Europe, this move has already started with the implementation of TKTD modelling into several risk assessment schemes (although the classical statistical methods are still in there as well).

It is good to note that this discussion is not just a matter of ERA: also for academic research, statistical dose-response analysis is not answering many interesting questions. Ignoring the factor of time, we cannot use summary statistics to compare the sensitivity of different species for the same chemical, nor the toxicity of different chemicals in the same species, and it does not increase our understanding of the underlying processes. Unfortunately, the statistical approaches are so embedded into ecotoxicology that their limitations are only rarely considered.

The current state-of-the-art TKTD models provide a useful starting point but need to be further tested and developed. In my opinion, progress in this area has been slow, research efforts have been rather fragmented, and funding opportunities are limited. This may relate to the multi-disciplinary nature of the topic: TKTD modelling sits at a crossroads of ecotoxicology, mathematics, statistics, chemistry and biology. However, it is about time that ecotoxicology gains a better understanding of toxic effects, and let's start by reclaiming ERA's extrapolation question from the statisticians. Statistical science is essential for ERA, but it is useless without linking it to meaningful process models.

## REFERENCES

- Alda Álvarez O, Jager T, Nuñez Coloa B, & Kammenga JE. 2006. Temporal dynamics of effect concentrations. *Environmental Science & Technology* 40, 2478-2484.
- Barsi A, Jager T, Collinet M, Lagadic L, & Ducrot V. 2014. Considerations for test design to accommodate energy-budget models in ecotoxicology: a case study for acetone in the pond snail *Lymnaea stagnalis*. *Environmental Toxicology and Chemistry* 33, 1466-1475.
- Bedaux JJM, & Kooijman SALM. 1994. Statistical analysis of bioassays based on hazard modelling. *Environmental and Ecological Statistics* 1, 303-314.

- EFSA. 2014. Scientific Opinion on good modelling practice in the context of mechanistic effect models for risk assessment of plant protection products. *EFSA journal* 12, 3589.
- EFSA. 2018. Scientific Opinion on the state of the art of Toxicokinetic/Toxicodynamic (TKTD) effect models for regulatory risk assessment of pesticides for aquatic organisms. *EFSA journal* 16, 5377.
- EFSA. 2023. Revised guidance on the risk assessment of plant protection products on bees (*Apis mellifera*, *Bombus* spp. and solitary bees). *EFSA Journal* 21, 7989.
- Jager T. 2013. All individuals are not created equal; accounting for interindividual variation in fitting life-history responses to toxicants. *Environmental Science & Technology* 47, 1664-1669.
- Jager T. 2014. Reconsidering sufficient and optimal test design in acute toxicity testing. *Ecotoxicology* 23, 38-44.
- Jager T. 2020. Revisiting simplified DEBtox models for analysing ecotoxicity data. *Ecological Modelling* 416, 108904.
- Jager T. 2024. Identifying and predicting delayed mortality with toxicokinetic–toxicodynamic models. *Environmental Toxicology and Chemistry* 43, 1030-1035.
- Jager T, Albert C, Preuss TG, & Ashauer R. 2011. General Unified Threshold model of Survival - a toxicokinetic-toxicodynamic framework for ecotoxicology. *Environmental Science & Technology* 45, 2529-2540.
- Jager T, & Ashauer R. 2018. How to evaluate the quality of toxicokinetic-toxicodynamic models in the context of environmental risk assessment. *Integrated Environmental Assessment and Management* 14, 604-614.
- Jager T, & Ashauer R. 2023. Modelling survival under chemical stress. A comprehensive guide to the GUTS framework. Toxicodynamics Ltd., York, UK. Available from Leanpub, [https://leanpub.com/guts\\_book](https://leanpub.com/guts_book), Version 3.0, 20 July 2023
- Kooijman SALM. 2018. Models in stress research. *Ecological Complexity* 34, 161-177.
- Kooijman SALM, & Bedaux JJM. 1996. Analysis of toxicity tests on *Daphnia* survival and reproduction. *Water Research* 30, 1711-1723.
- Marques GM, Lika K, Augustine S, Pecquerie L, & Kooijman SALM. 2019. Fitting multiple models to multiple data sets. *Journal of Sea Research* 143, 48-56.
- OECD. 2006. Current approaches in the statistical analysis of ecotoxicity data: a guidance to application, Series on Testing and Assessment, No. 54. Organisation for Economic Cooperation and Development (OECD), Paris, France
- Sherborne N, Galic N, & Ashauer R. 2020. Sublethal effect modelling for environmental risk assessment of chemicals: problem definition, model variants, application and challenges. *Science of The Total Environment* 745, 141027.
- Sprague JB. 1969. Measurement of pollutant toxicity to fish. I. Bioassay methods for acute toxicity. *Water Research* 3, 793-821.